

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วน โดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 4 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

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Mahidol University
Faculty of Information
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ITDS261 Introduction to Software Engineering

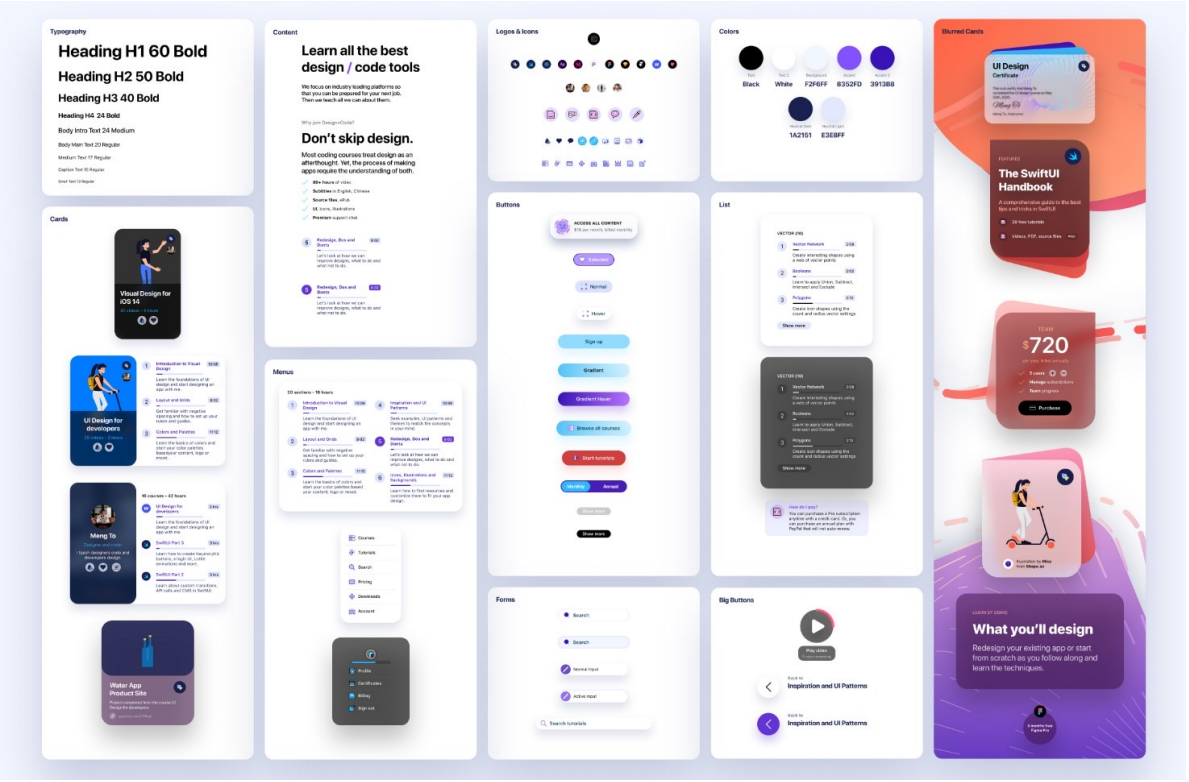
Lecture 8 – User Interface Design

Topics

- User Interface Design
  - Good user interface design guidelines
  - Use of UI design prototype

User Interface (UI)

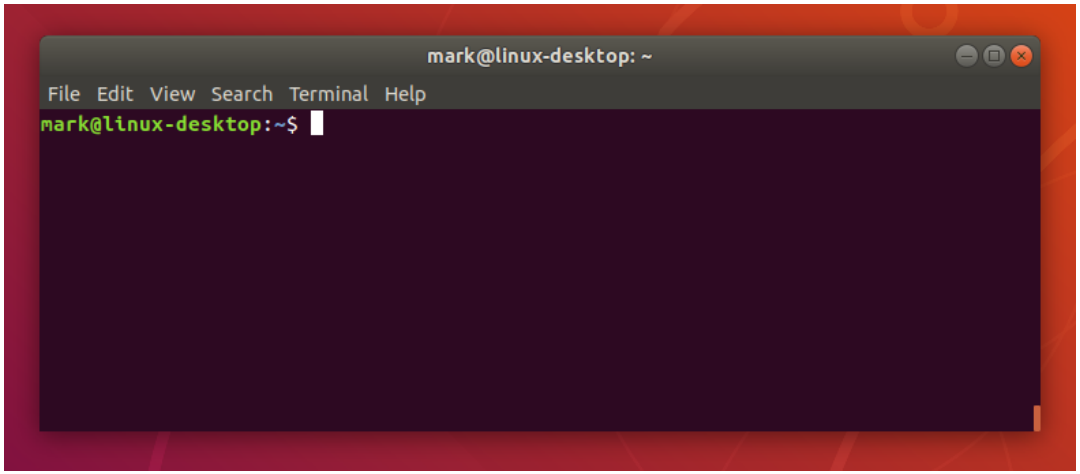
- Most apparent to the user
- Two main issues
  1. Flow of interactions
  2. Look and feel



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Types of User Interfaces

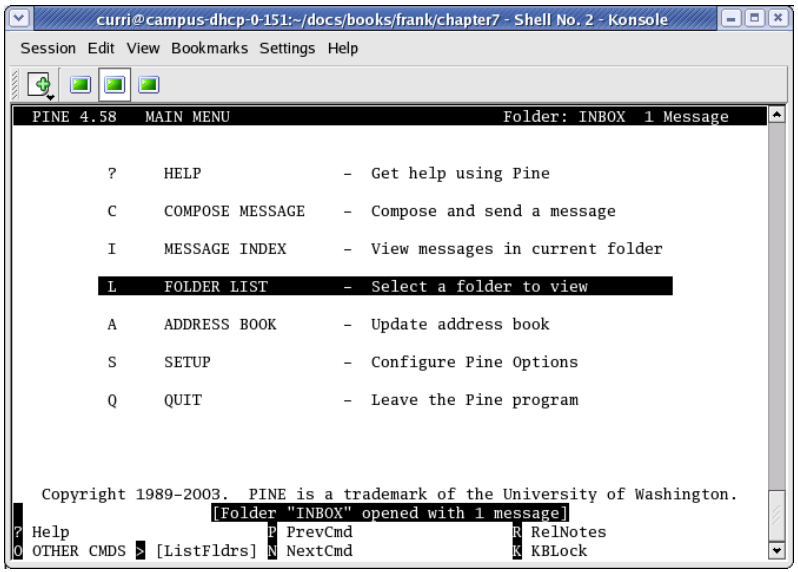
- **Command-Line Interfaces (CLI):** Relies on text-based commands for user interaction, often used by developers and system administrators for precise control over operations.



Reference: https://dapth.com/insights/user-interface-ui

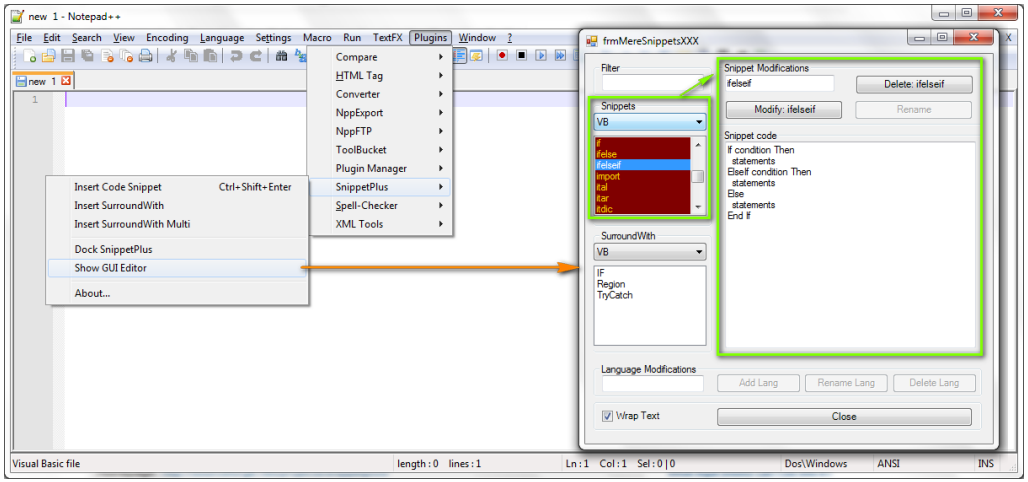
Types of User Interfaces (Cont.)

- **Text Menu Interfaces:** A type of text-based user interface; Displays menus as text on the screen. These menus guide the user to select options, usually by pressing numbers, letters, or arrow keys.



Types of User Interfaces

- **Graphical User Interfaces (GUI):** Uses graphical elements like icons and windows to facilitate interaction, commonly found in operating systems and applications.





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Reference: https://dapth.com/insights/user-interface-ui

Types of User Interfaces (Cont.)

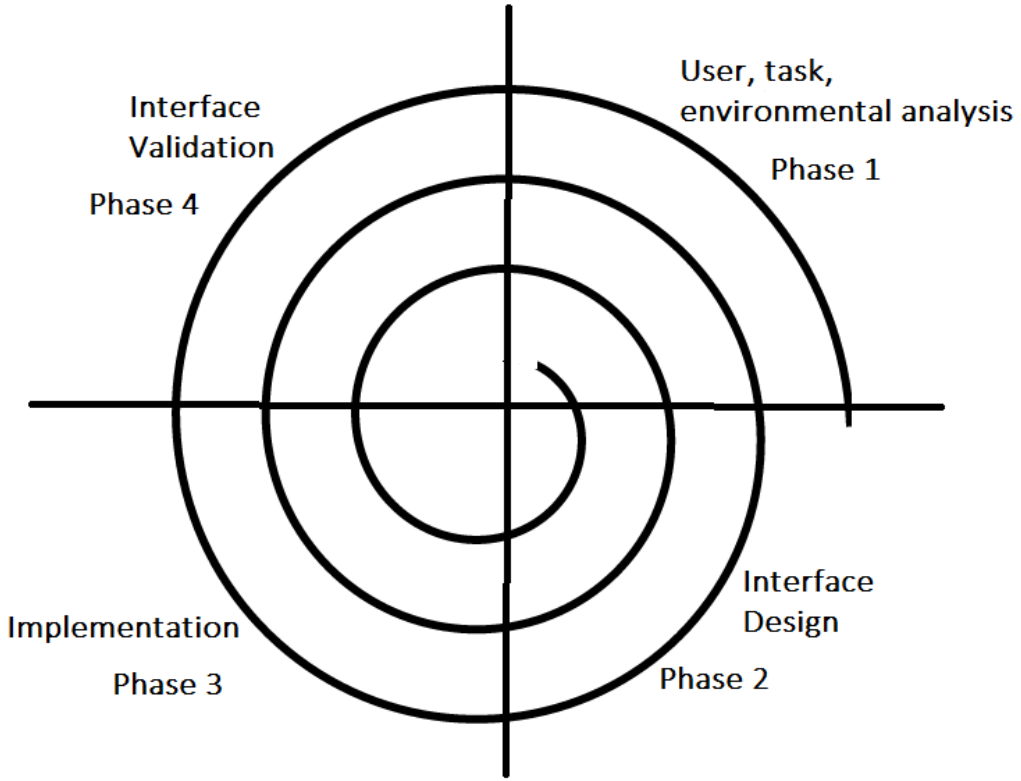
- **Touchscreen User Interfaces:** Allows users to interact with devices through touch gestures, prevalent in smartphones, tablets, and interactive kiosks.
- **Voice User Interfaces (VUI):** Enables interaction through spoken commands and responses, popularised by virtual assistants like Siri and Alexa.
  - Example: https://www.youtube.com/watch?v=eePfgI2TT_Q
- **Gesture-Based Interfaces:**
  - **Augmented Reality (AR) User Interfaces:** Integrates virtual elements into the natural environment, commonly seen in AR apps for navigation, gaming, and education.
  - **Virtual Reality (VR) User Interfaces:** Creates immersive virtual environments for user interaction, utilised in VR gaming, simulations, and training programs.
  - Example: https://www.youtube.com/watch?v=5_bVkbG1ZCo

Basic Rules

The software becomes more popular if its user interface is:

- Attractive
- Simple to use
- Responsive in short time
- Clear to understand
- Consistent on all interface screens

User Interface Design Process



User Interface Design Process

- **Phase 1: User, task, environmental analysis, and modeling**
  - Based on the profile of user
    - Understand skill, knowledge, and type of user
  - Tasks that user need to perform
  - Physical work environment

User Interface Design Process

- Example questions:
  - Where will the interface be located physically?
  - Will the user be sitting, standing, or performing other tasks unrelated to the interface?
  - Does the interface hardware accommodate space, light, or noise constraints?
  - Are there special human factors considerations driven by environmental factors?

User Interface Design Process







User Interface Design Process

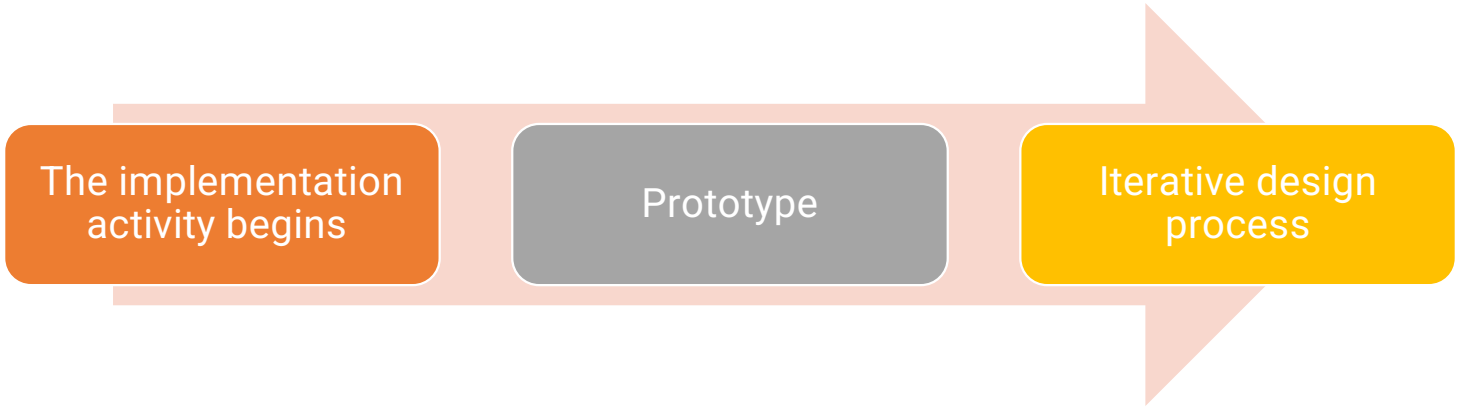
Phase 2: Interface Design

- Set of interface objects
- Set of interface actions
- Design issues
  - Response time
  - Command and action structure
  - Error handling



User Interface Design Process

- **Phase 3: Interface construction and implementation**



User Interface Design Process

- **Phase 4: Interface Validation**
  - Testing the interface
    - Can implement every user task correctly
    - Can accommodate all task variations
    - Achieve all general user requirements
  - Assessing the degree to which the interface is easy to use and easy to learn
  - Users should accept the interface as a useful tool in their work

Golden Rules for Interface Design by Theo Mandel

Three golden rules by Theo Mandel:

- Place the user in control
- Reduce the user's memory load
- Make the interface consistent

https://theomandel.com/resources/golden-rules-of-user-interface-design/

Golden Rules (Cont.)

- Place the user in control
  - Define interaction modes in a way that does not force a user into unnecessary or undesired actions
    - If *spell check* mode is selected in a document editor, the screen should show a spell-checking mode. Also allow a user to exit this mode easily.
  - Provide for flexible interaction
    - Allow a user to interact via keyboard commands, mouse movement, a touch screen or by voices
  - Allow user interaction to be interruptible and undoable
  - Allow advance users to customize their interface to facilitate interaction
  - Hide technical interfaces from casual users
  - Design for direct interaction with objects that appear on the screen
    - Allow a user to drag a document file to a “trash” icon – *deleting a file*

Golden Rules (Cont.)

- Reduce the user's memory load
  - Reduce demand on short-term memory
    - Provide visual clues that enable user to recognize past actions
  - Establish meaningful defaults
    - Initial set of defaults should make sense for the average user
  - Define shortcuts that are intuitive
    - Ctrl + P for a print command, Ctrl + S for a save command, Cmd + C for a copy command
  - Visual layout of the interface should be based on a real-world metaphor
    - Use a *house* symbol for "Back to home screen" command
  - Disclose information in layers
    - Display general information first, followed by additional details

Golden Rules (Cont.)

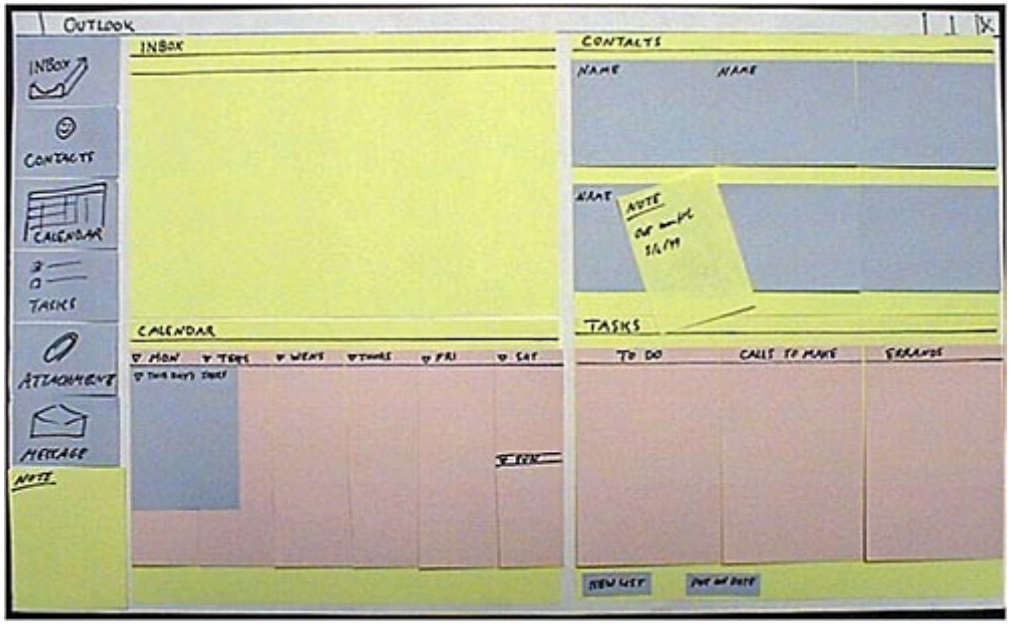
- Make the interface consistent
  - Allow a user to put the current task into a meaningful context
    - Provide *indicators* (e.g., window titles, graphical icons and consistent color coding) that enable the user to know the context of the work at hand
  - Maintain consistency across the system
  - Do not make changes to the existing interactive models
    - Ctrl + S is well known for saving files – *do not use* the same shortcut to execute another command

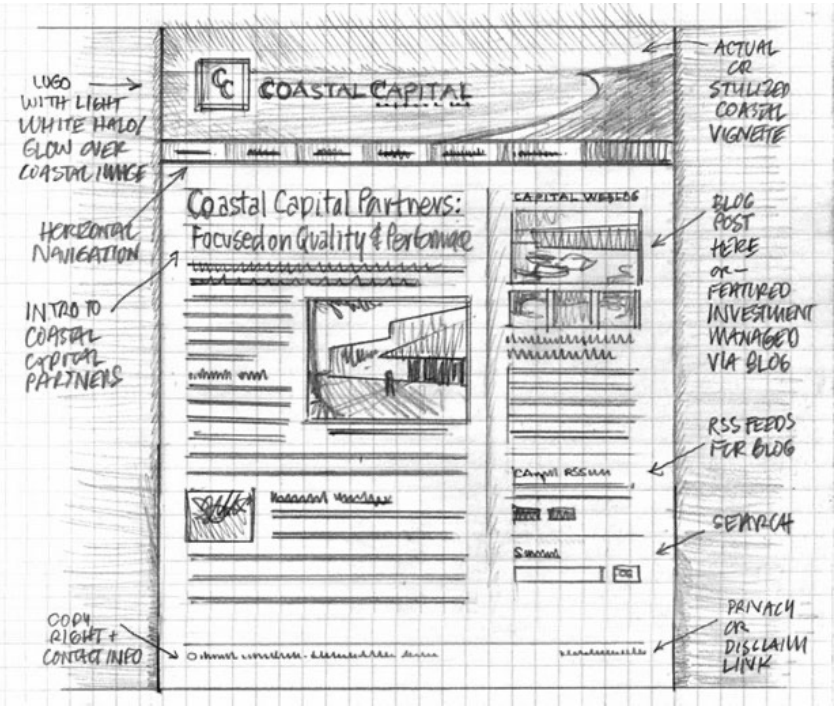
UI Design Prototype

- There are two types of UI design prototypes:
  - Low fidelity
    - Simple and rough
    - Focus on basic structure and functionality, avoiding design details
    - Useful for early-stage feedback and brainstorming
    - E.g., Paper prototypes, click-through prototypes
  - High fidelity
    - Highly detailed, with realistic UI elements, content, and interaction
    - Mimics the final product as closely as possible, sometimes including animations and transitions
    - Allows for accurate user testing and demonstration of final design
    - Useful for validating design, usability, and functionality before development
    - E.g., Digital, code free prototypes, coded prototypes

https://www.protopie.io/blog/low-fidelity-vs-high-fidelity-prototyping

Low Fidelity Prototypes

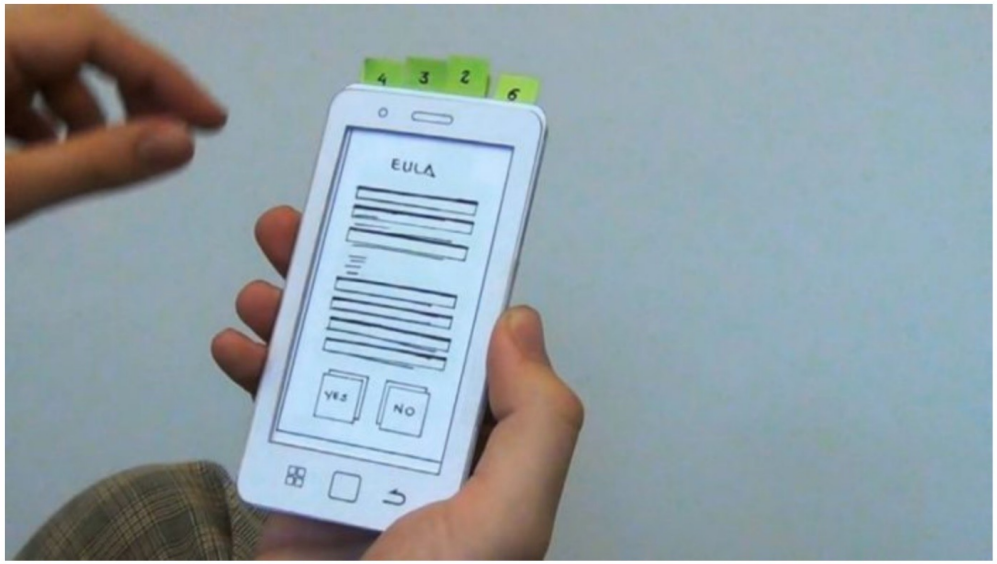




Low Fidelity Prototypes (Cont.)

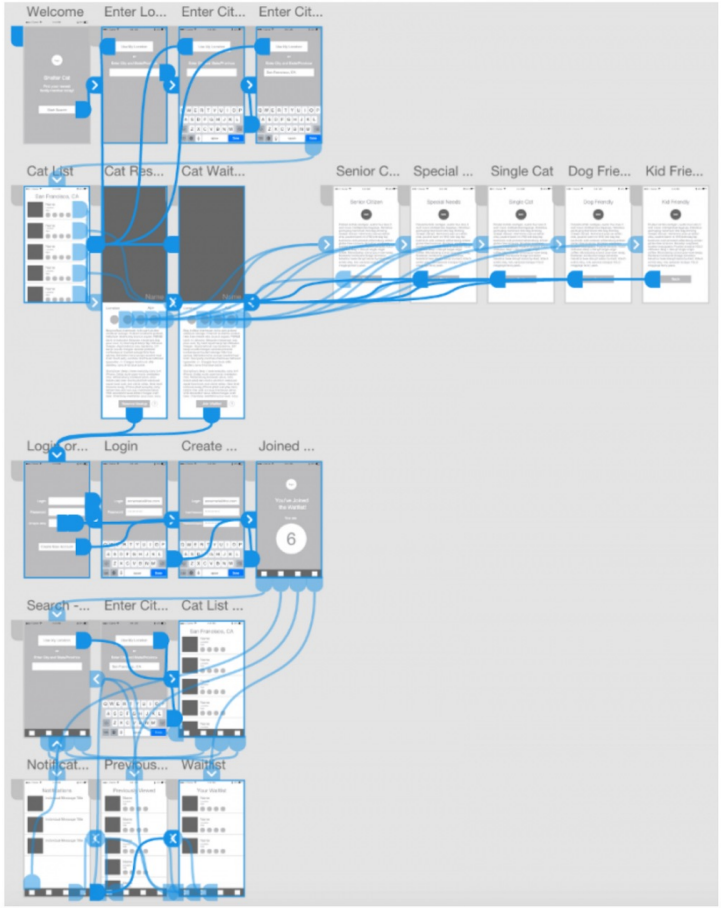


Simple screens are drawn on paper. A common practice for testing these prototypes is to have one person play 'computer,' switching the sketches around according to user choices. Image source: UX Playground



User interfaces can be quickly prototyped and tested. Image source: Vimeo

Low Fidelity Prototypes (Cont.)



An example of a low-fidelity prototype made in Adobe XD.

High Fidelity Prototypes

